

1970 British Cohort Study

Response and Deaths data

User Guide (Version 6)

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CENTRE FOR
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STUDIES



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About the 1970 British Cohort Study

The 1970 British Cohort Study (BCS70) is an ongoing, multi-disciplinary longitudinal birth cohort study, following a nationally representative sample of over 16,000 people born in Britain in a single week in April 1970.

We have surveyed cohort members throughout their childhood and adult lives, mapping their individual trajectories and creating a unique resource for researchers. It is one of very few longitudinal studies following people of this generation anywhere in the world.

Featuring a range of objective measures and rich self-reported data, BCS70 covers an incredible amount of ground and can be used in research on many topics

Evidence from BCS70 has illuminated important issues for our society across five decades. Key findings include how reading for pleasure matters for children's cognitive development, why grammar schools have not reduced social inequalities, and how childhood experiences can impact on mental health in mid-life.

Every day researchers from across the scientific community are using this important study to make new connections and discoveries.

1. Introduction

This document accompanies the release of the sixth edition of the BCS70 Response and Deaths data at the UK Data Service. The new data replace the previous response datasets and combines it with data on known deaths among members of the British Cohort Study birth cohort from 1970 to 2024.

The BCS70 study began as the British Births Survey, when data were collected over babies born in Great Britain (England, Scotland and Wales) in one week of 1970. To date, there have been nine follow-ups of the entire cohort which have sought to monitor their development in key domains – health, education, social and economic.

The follow-ups were undertaken when the cohort members were aged 5, 10, 16, 26, 30, 34, 38, 42, 46 and 51 years. For the first three of these, the birth cohort was augmented by including in the target sample immigrants born in the relevant week and identified from school registers during tracing. Sub-studies have also been carried out at age 22 months, 42 months and 21 years. Over the years, information has been gathered from a number of sources (e.g. parents, schools, doctors, medical records, cohort members) and in a variety of ways (e.g. interview, self-completion, assessments, medical records). In addition, details of public examination entry and performance were obtained from schools prior to the 26-year follow-up.

2. Survey Response

In this section we describe the ‘cross-sectional’ and ‘longitudinal’ response rates achieved at each follow-up. This builds on the work originally done by Plewis et al¹ which described response up to 2004. The response dataset has now been updated with data from the most recent survey – BCS70 Ahe 51, Sweep 11 (2021).

The birth sample includes approximately 98% of all notified births in Great Britain (England, Wales and Scotland) during one week of April 1970. At BCS2 (1975), BCS3 (1980) and BCS4 (1986) immigrants to Great Britain who were born in the

¹ PLEWIS, I, CALDERWOOD, L, HAWKES, D AND NATHAN, G (2004) [*Changes in the NCDS and BCS70 populations and samples over time*](#). CLS Technical Report. London: CLS, Institute of Education

relevant week were identified from school registers and added to the sample².

Additional children born in Great Britain not previously contacted for the study during the week in question were also added.

The first sweep of BCS70 included 628 cases from Northern Ireland which were never followed up. While their data remains available to access, it has been removed from the latest version of the response data as the focus is on longitudinal response. A further 341 cases have also been dropped from the dataset having never participated in any of the 11 sweeps.

Table 1 shows how many sweeps cohort members have participated in BCS70. 64% have completed 6 or more sweeps of data collection.

Table 1: Completed sweeps

Number of sweeps completed	Number of cohort members	Percentage	Cumulative percentage
11	2763	15%	15%
10	2410	13%	29%
9	1910	11%	40%
8	1522	8%	48%
7	1315	7%	56%
6	1380	8%	64%
5	1356	8%	67%
4	1571	9%	78%
3	1209	7%	85%
2	813	5%	90%
1	1789	10%	100%
Total	18038		

Table 2 shows a summary of the survey responses at every sweep, with **Table 2a** displaying these number of percentage terms (%). It indicates the number of productive interviews achieved in each follow-up from all cases, whether born in

² FERRI, E, BYNNER, J and WADSWORTH, M. (2003) Changing Britain, Changing Lives: Three Generations at the End of the Century. London: Institute of Education.

Great Britain or otherwise. Where productive interviews were not achieved, the reason for this is also shown. The category “not issued” has a different context prior to BCS4 and from BCS5 onwards. In the first three sweeps “not issued” are those cohort members who joined the study after the birth data collection, either because they were immigrants or were missed during the initial data collection in 1970. From BCS5 onwards the ‘not issued’ category is comprised of those who chose to withdraw permanently from the study and those who had been untraced for a lengthy period. Guidance on how to restore sample representativeness despite non-response is available in the CLS Missing Data User Guide.

Table 3 shows the number of cohort members resident in Great Britain by country of birth. **Appendix A** show a more detailed breakdown of response by country of birth and residence.

Table 4 shows the calculated cross-sectional and longitudinal samples and consequent response rates using information from Table 2:

1. The cross-sectional sample at sweep *t* is defined as “*all those born during the defined week in April 1970, and living in Great Britain at the time of the sweep *t* data collection (including the Isle of Man and Channel Islands)*”.
2. The cross-sectional achieved sample is “all those members of the cross-sectional sample who participate in a particular sweep.”
3. The longitudinal sample is defined as “*all those born in Great Britain during the defined week in April 1970, and living in Great Britain at the time of the sweep *t* data collection (including the Isle of Man and Channel Islands)*”.
4. The longitudinal achieved sample is “all those members of the longitudinal sample who participate in a particular sweep.”

At the most recent sweep both the cross-sectional and longitudinal response rates were 70%. The percentage of the cross-sectional and longitudinal samples is the same in each sweep. The response rate was handicapped in 1986 by a teachers’ strike and in 1996 by the use of a postal survey with limited resources.

Table 2: Survey Response

	BCS1	BCS2	BCS3	BCS4	BCS5	BCS6	BCS7	BCS8	BCS9	BCS10	BCS11
	<i>Birth</i>	<i>Age 5</i>	<i>Age 10</i>	<i>Age 16</i>	<i>Age 26</i>	<i>Age 30</i>	<i>Age 34</i>	<i>Age 38</i>	<i>Age 42</i>	<i>Age 46</i>	<i>Age 51</i>
<i>OUTCOME</i>	1970	1975	1980	1986	1996	2000	2004	2008	2012	2016	2021
Productive	16589	13135	14870	11615	9003	11261	9665	8874	9841	8581	8016
Refusal	0	0	0	0	66	1080	1002	710	1864	2068	2609
Non-contact	0	0	0	5773	4163	2681	2003	1926	1212	1120	1368
Other unproductive	17	3257	2340	18	47	459	231	199	340	415	391
Ineligible	0	0	0	0	0	15	1	4	0	8	40
Not Issued	1432	1079	239	10	4008	1541	3885	4988	3383	4395	4325
Not Issued - Emigrant	0	0	0	0	35	235	432	456	433	466	242
Dead	0	567	589	622	716	766	819	881	965	985	1047
Total	18038	18038	18038	18038	18038	18038	18038	18038	18038	18038	18038

Table 2a: Survey Response (%)

	BCS1	BCS2	BCS3	BCS4	BCS5	BCS6	BCS7	BCS8	BCS9	BCS10	BCS11
	<i>Birth</i>	<i>Age 5</i>	<i>Age 10</i>	<i>Age 16</i>	<i>Age 26</i>	<i>Age 30</i>	<i>Age 34</i>	<i>Age 38</i>	<i>Age 42</i>	<i>Age 46</i>	<i>Age 51</i>
<i>OUTCOME</i>	1970	1975	1980	1986	1996	2000	2004	2008	2012	2016	2021
Productive	92%	73%	82%	64%	50%	62%	54%	49%	55%	48%	44%
Refusal	0%	0%	0%	0%	0%	6%	6%	4%	10%	11%	15%
Non-contact	0%	0%	0%	32%	23%	15%	11%	11%	7%	6%	8%
Other unproductive	0%	18%	13%	0%	0%	3%	1%	1%	2%	2%	2%
Ineligible	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Not Issued	8%	6%	1%	0%	22%	9%	22%	28%	19%	24%	24%
Not Issued - Emigrant	0%	0%	0%	0%	0%	1%	2%	3%	2%	3%	1%
Dead	0%	3%	3%	4%	4%	4%	5%	5%	5%	6%	6%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Table 3: Residence/Country of birth at each follow-up

	BCS1	BCS2	BCS3	BCS4	BCS5	BCS6	BCS7	BCS8	BCS9	BCS10	BCS11
	<i>Birth</i>	<i>Age 5</i>	<i>Age 10</i>	<i>Age 16</i>	<i>Age 26</i>	<i>Age 30</i>	<i>Age 34</i>	<i>Age 38</i>	<i>Age 42</i>	<i>Age 46</i>	<i>Age 51</i>
	1970	1975	1980	1986	1996	2000	2004	2008	2012	2016	2021
Resident in Great Britain	17566	17082	17376	17416	17287	17037	16787	16701	16640	16587	16522
- born in Great Britain	17566	16999	16977	16944	16817	16588	16348	16265	16206	16154	16088
- born outside of Great Britain*	0	83	399	472	470	449	439	436	434	433	434
Not resident in Great Britain	472	389	73	0	35	235	432	456	433	466	469
- born in Great Britain	0	0	0	0	34	216	405	428	406	438	442
- born outside of Great Britain*	472	389	73	0	1	19	27	28	27	28	27
Died	0	567	589	622	716	766	819	881	965	985	1047
Total	18038	18038	18038	18038	18038	18038	18038	18038	18038	18038	18038

*Includes 26 cohort members where country of birth not known.

Table 4: Target and achieved samples

	BCS1	BCS2	BCS3	BCS4	BCS5	BCS6	BCS7	BCS8	BCS9	BCS10	BCS11
	<i>Birth</i>	<i>Age 5</i>	<i>Age 10</i>	<i>Age 16</i>	<i>Age 26</i>	<i>Age 30</i>	<i>Age 34</i>	<i>Age 38</i>	<i>Age 42</i>	<i>Age 46</i>	<i>Age 51</i>
	1970	1975	1980	1986	1996	2000	2004	2008	2012	2016	2021
Cross-sectional target sample*	16606	16390	17207	17405	13279	15494	12900	11711	13255	12190	12424
Cross-sectional achieved sample	16589	13135	14867	11615	9003	11261	9665	8874	9841	8581	8016
Cross-sectional achieved sample (%)	100%	80%	86%	67%	68%	73%	75%	76%	74%	70%	65%
Longitudinal target sample**	16585	16307	16808	16933	12958	15081	12590	11442	12936	11897	12283
Longitudinal achieved sample	16568	13052	14498	11305	8795	11004	9449	8671	9615	8376	7825
Longitudinal achieved sample (%)	100%	80%	86%	67%	68%	73%	75%	76%	74%	70%	64%

* All those known to be alive and in the issued sample

** All those known to be alive born in Great Britain and in the issued sample

3. Deaths

Information on deaths has been taken from the records maintained by the organisations responsible for the study over the life time of the study - the National Birthday Trust Fund, the National Children's Bureau (NCB), the Social Statistics Research Unit (SSRU) and the Centre for Longitudinal Studies (CLS). The information has been gleaned from a variety of sources, including death certificates and other information from the NHS.

Central Register (NHSCR), and from relatives and friends during survey activities and cohort maintenance work by telephone, letter and e-mail. Records have not been updated for the 2016 sweep so currently death data is only available to 2012.

Sources of deaths data

- *NHSCR Death Certificates*: death certificates supplied to SSRU/CLS by the NHSCR
- *PMS/BCS70 Death Cards*: a card index which recorded deaths among the BCS70 cohort from 1970 until (approximately) 1981.
- *Survey/Cohort Maintenance work*: information deriving from survey activities (e.g. interviewers being told on the doorstep that cohort member has died), or from the BCS70 Address Database, compiled from cohort maintenance work (e.g. response to feedback, phone calls and correspondence between surveys).
- *Stillbirth/Neonatal Death*: these deaths occurred during the period when the BCS70 interviewers were still in close contact with the family, and so verification was immediate.

Diagram 1 provides information on death by sex. This chart shows the expected higher mortality of males, especially after the age of sixteen.

Diagram 2 shows the higher mortality in Scotland and Wales, compared to England.

Diagram 1: Cumulative deaths by sex (up to 2012)

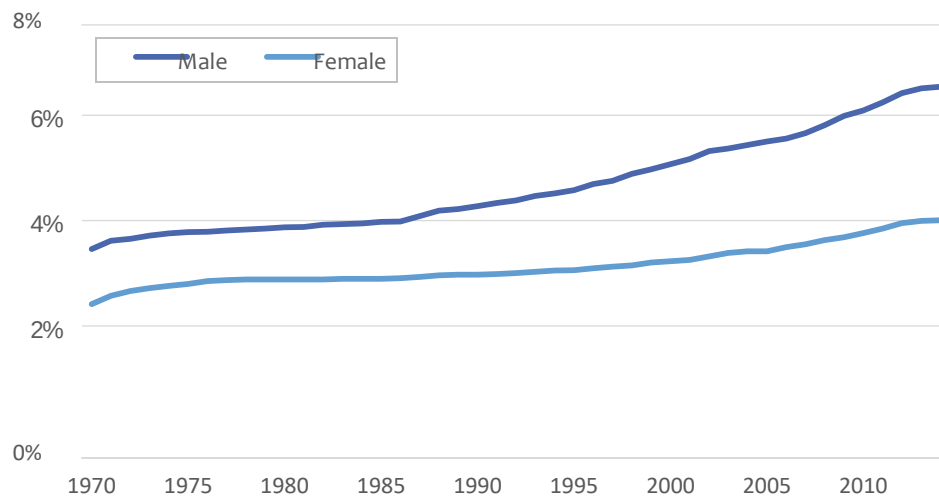
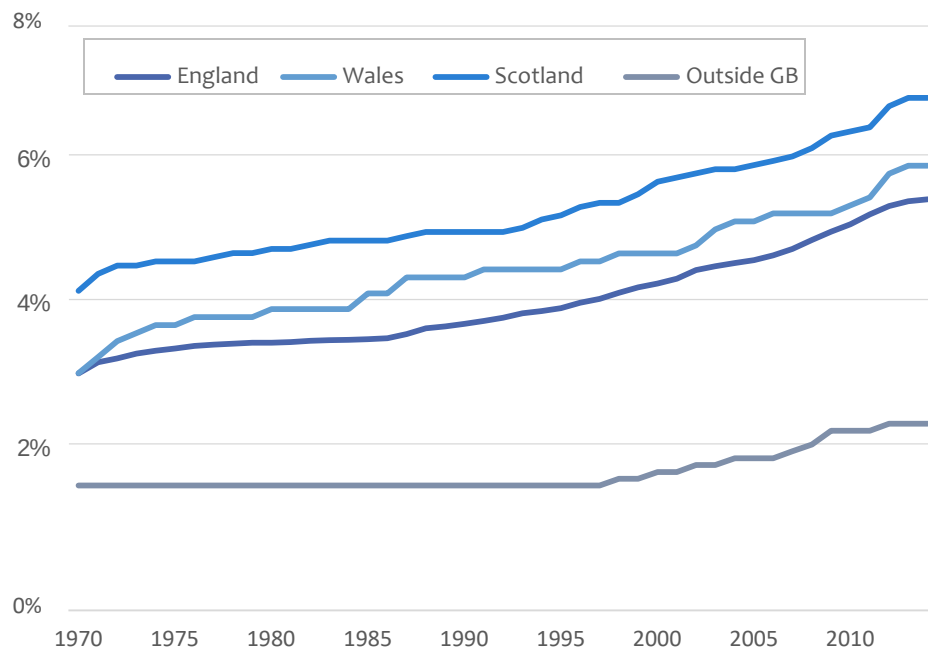


Diagram 2: Cumulative deaths by country of birth (up to 2012)



4. Research Data

4.1 Licencing and data access

The response data have been processed by CLS and supplied to the UK Data Service. All data users need to be registered with the UK Data Service and to sign the UKDS End User Licence. Details of how to do this are available at ukdataservice.ac.uk/get-data/how-to-access/registration.

The outcomes for sweeps 1-11 are available from the UK Data Service as safeguarded data, which can be downloaded from the UK Data Service once the End User Licence (EUL) access conditions have been accepted by the user.

Mortality data are available as special safeguarded data at the UK Data Service. These datasets can be requested using the UK Data Service Special Licence application form. Once the form has been reviewed by UK Data Service and approved by the CLS Data Access Committee the data will be available to download.

4.2 List of Datasets

Dataset name	Content summary
bcs70_response_1970-2021	Overall response file containing outcomes for all main survey sweeps
bcs_outcome_death_protect	Details of deaths for cohort members who have died (up to 2012)

4.3 Data Documentation

The response data deposit includes additional historical documentation not maintained for recent data changes but still made available for users

Table 8: Data documents

Name of the document	Content summary
ncds_and_bcs70_response.pdf	Report on changes in sample numbers over time (last updated 2004)

4.4 Dataset Description

Survey response dataset (End User Licence)

The content of the EUL dataset is summarised in the **Table 5**.

Table 5: Response data variables

Variable	Description
BCSID	Research Serial Number
SEX	Sex of cohort member
COB	Country of Birth
MULTIPNO	Twin identity
TWINCODE	Corresponding twin identifier
OUTCME01	Outcome to BCS1 (1970)
OUTCME02	Outcome to BCS2 (1975)
OUTCME03	Outcome to BCS3 (1980)
OUTCME04	Outcome to BCS4 (1986)
OUTCME05	Outcome to BCS5 (1996)
OUTCME06	Outcome to BCS6 (2000)
OUTCME07	Outcome to BCS7 (2004)
OUTCME08	Outcome to BCS8 (2008)
OUTCME09	Outcome to BCS9 (2012)

Variable	Description
OUTCME10	Outcome to BCS10 (2016)
OUTCME11	Outcome to BCS11 (2021)

Value labels

Outcomes are categorised with the following codes:

0. Refusal/Non-contact
1. Productive
2. Refusal
3. Non-contact
4. Other unproductive
5. Ineligible
6. Not issued
7. Not issued – Emigrant
8. Dead

Death dataset (Special Licence)

The content of the death dataset is summarised in the **Table 7**

Table 7: Deaths data variables

Variable	Description
BCSID	Research Serial Number
DODMTH	Consolidated month of death
DODYR	Consolidated year of death
DSOURCE	Source of death information

Appendix A: Detailed Response by country of birth and residence

	BCS1 <i>Birth</i> 1970		BCS2 <i>Age 5</i> 1975		BCS3 <i>Age 10</i> 1980		BCS4 <i>Age 16</i> 1986		BCS5 <i>Age 26</i> 1996		BCS6 <i>Age 30</i> 2000		BCS7 <i>Age 34</i> 2004		BCS8 <i>Age 38</i> 2008		BCS9 <i>Age 42</i> 2012		BCS10 <i>Age 46</i> 2016		BCS11 <i>Age 51</i> 2021	
OUTCOME	<i>GB</i>	<i>Not GB</i>	<i>GB</i>	<i>Not GB</i>	<i>GB</i>	<i>Not GB</i>	<i>GB</i>	<i>Not GB</i>	<i>GB</i>	<i>Not GB</i>	<i>GB</i>	<i>Not GB</i>	<i>GB</i>	<i>Not GB</i>	<i>GB</i>	<i>Not GB</i>	<i>GB</i>	<i>Not GB</i>	<i>GB</i>	<i>Not GB</i>	<i>GB</i>	<i>Not GB</i>
Productive	16568	21	13052	83	14501	369	11305	310	8796	207	11004	257	9449	216	8671	203	9615	226	8376	205	7825	191
Refusal	0	0	0	0	0	0	0	0	64	2	1055	25	985	17	693	17	1827	37	2021	47	2546	63
Non-contact	0	0	0	0	0	0	5623	150	4054	109	2578	103	1931	72	1883	43	1175	37	1090	30	1311	57
Other unproductive	17	0	3257	0	2310	30	6	12	45	2	432	27	226	5	193	6	321	19	404	11	385	6
Ineligible	0	0	0	0	0	0	0	0	0	0	14	1	1	0	4	0	0	0	8	0	36	4
Not Issued	981	451	690	389	166	73	10	0	3858	150	1505	36	3756	129	4821	167	3268	115	4255	140	4201	124
Not Issued - Emigrant	0	0	0	0	0	0	0	0	34	1	216	19	405	27	428	28	406	27	438	28	226	16
Dead	0	0	567	0	589	0	622	0	715	1	762	4	813	6	873	8	954	11	974	11	1036	11
Resident	17566	0	16999	83	16977	399	16944	472	16817	470	16588	449	16348	439	16265	436	16206	434	16154	433	16088	434
Not resident	0	472	0	389	0	73	0	0	34	1	216	19	405	27	428	28	406	27	438	28	442*	27*
Dead	0	0	567	0	589	0	622	0	715	1	762	4	813	6	873	8	954	11	974	11	1036	11
Total	17566	472	17566	472	17566	472	17566	472	17566	472	17566	472	17566	472	17566	472	17566	472	17566	472	17566	472
Total by data collection	18038		18038		18038		18038		18038		18038		18038		18038		18038		18038		18038	

*Includes 227 (216 GB +11 Not GB) productive mop-up cases